

# Prajwal Sah

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**Builder by instinct | Engineer by discipline | Poet by passion**

I build across the boundaries of software engineering, AI/ML, and cybersecurity – and lead technical communities as a student ambassador.

## Education

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| <b>Khwopa Engineering College, Bhaktapur</b><br><i>Bachelor of Engineering in Computer Engineering</i> | 2023 – Present<br><i>Expected Graduation: Mar 2027   CGPA: 3.70/4.0</i> |
| <b>Khwopa Higher Secondary School, Bhaktapur</b><br><i>Higher Secondary Education (+2 Science)</i>     | 2022<br><i>GPA: 3.50</i>                                                |
| <b>Sainik Awashiya Mahavidyalaya, Bhaktapur</b><br><i>Secondary Education Examination (SEE)</i>        | 2020<br><i>GPA: 3.80</i>                                                |

## Experience

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| <b>Vortex Tech</b><br><i>Cybersecurity Intern (Project-Based)</i> | Jul 2026 – Aug 2026<br><i>Remote</i> |
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- Identified open ports, active services, and potential attack surfaces across test environments, documenting results in a structured security assessment report, by conducting Nmap-based network reconnaissance and vulnerability scanning.
- Improved end-user password security awareness, translating findings into actionable recommendations, by developing a password strength checker that evaluated key password characteristics.
- Built foundational offensive-security competency, validated through weekly technical assignments and documentation, by applying Linux, networking, and reconnaissance best practices in hands-on exercises.

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| <b>Bhaktapur Municipality (Collaboration through Khwopa Engineering College)</b><br><i>Software Developer</i> | Jan 2026 – May 2026<br><i><a href="#">Live Website</a></i> |
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- Delivered a full-stack tourism and digital heritage platform officially adopted and publicly hosted by Bhaktapur Municipality, by building **15+ responsive pages** and **15+ service modules** with Django, HTML, CSS, JavaScript, Bootstrap, and PostgreSQL.
- Enabled location-aware discovery of nearby attractions with configurable search radii, powering interactive maps for platform users, by implementing secure authentication and **REST API** integrations for geolocation features.
- Shipped the platform across the full development lifecycle within a **4-member team**, meeting municipal deployment requirements, by leading requirement analysis, implementation, testing, and stakeholder feedback cycles.
- Achieved production readiness ahead of public launch, resulting in official municipal adoption, by owning backend integration, database design, and deployment.

## Research and Ongoing Works

- Developing LSA-Agent, a privacy-preserving Agentic RAG framework, to automate security incident investigation by combining Wazuh SIEM data with local LLMs.
- Researching AI-assisted threat hunting and on-premise cybersecurity analytics by mapping detection workflows to the MITRE ATT&CK framework in privacy-preserving environments.
- Evaluating local large language models for cybersecurity incident analysis and alert triage to assess their effectiveness against cloud-based alternatives.

## Technical Skills

**Languages:** Python, C, C++, JavaScript, HTML, CSS, ReactJS (Workshop/Practical Exposure)

**Data Structures & Algorithms:** Arrays, Linked Lists, Trees, Graphs, Dynamic Programming

**Frameworks & Libraries:** Django, Flask, REST APIs, Bootstrap, Tailwind CSS

**AI & Machine Learning:** LLM APIs, Retrieval-Augmented Generation (RAG), Agentic AI, NLP, Prompt Engineering

**Databases:** PostgreSQL, MySQL, MongoDB

**Cybersecurity Tools:** Kali Linux, Wazuh SIEM, MITRE ATT&CK, Wireshark, Nmap

**Tools & Fundamentals:** Git, GitHub, LaTeX, WordPress, Agile/Scrum, Testing & Debugging, Problem Solving

## Projects

**Fallacy Detector & AI Reasoning Assistant**

*KU Hackathon 2025 – AI/ML Track Winner*

*[Github](#)*

- Achieved **82% classification accuracy** across **13 fallacy categories** on **1,000+ annotated arguments**, by fine-tuning an ELECTRA transformer model integrated with **OpenRouter** LLM APIs.
- Delivered real-time reasoning assistance directly in the browser, supporting analysis of both selected webpage text and user-authored content, by building a **Manifest V3** browser extension.
- Won the **AI/ML Track** at KU Hackathon 2025 against university-wide competition, by architecting, developing, and presenting the solution within a **4-member team** over a **48-hour** hackathon.

## Spotify Clone Web Application

*Personal Project*

*Github*

- Built a **10-page** full-stack music streaming application replicating core Spotify functionality, by implementing **Django**-backed authentication, playlist CRUD operations, and audio playback.

## Repository Management System

*Personal Project*

*Github*

- Automated repository and file-state lifecycle management with complete CRUD operations, by engineering a **Python/PostgreSQL** system on a modular relational architecture.

## Canteen Management System

*Team Project*

*Github*

- Streamlined canteen operations end-to-end, covering login, billing, inventory, and receipt generation, by building a GUI-based desktop application in **C++** with **graphics.h** within a **4-member team**.

## Achievements

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- **Winner – AI/ML Track, KU Hackfest 2025** [Certificate]
- **Student Ambassador – GeeksForGeeks Campus Mantri 2026** [Certificate]
- **Student Partner – Internshala, ISP-2026** [Certificate]
- **2nd Runner-Up – Hult Prize at KhEC 2023** [Certificate]

## Leadership and Extracurricular

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- Strengthened cross-functional delivery on Agile/Scrum teams, spanning UI/UX, backend integration, and project coordination responsibilities.
- Tackled real-world security vulnerabilities in the Pentester Nepal PMO-Sponsored Challenge, applying practical penetration testing methodologies to solve complex cybersecurity problems.
- Competed in the NCA Hack The Box CTF Challenge, successfully capturing both the **user** and **root** flags by troubleshooting systems and exploiting infrastructure vulnerabilities under time constraints.
- Sharpened collaborative problem-solving under real-world constraints, by actively participating in hackathons, ideathons, and competitive technical challenges.
- Served as **Vice President**, IT Circle – Khwopa Engineering College, leading committee initiatives and coordinating technical event planning.
- Produced promotional materials and visual assets for college technical events and club activities, as **Graphic Designer** for IT Circle, Khwopa Engineering College.